

ChatGPT tells stories, and a note about reverse engineering A Working Paper

William Benzon

March 3, 2023

Abstract: I examine a set of stories that are organized on three levels: 1) the entire story trajectory, 2) segments within the trajectory, and 3) sentences within individual segments. I conjecture that the probability distribution from which ChatGPT draws next tokens follows a hierarchy nested according to those three levels and that is encoded in the weights off ChatGPT's parameters. I arrived at this conjecture to account for the results of experiments in which ChatGPT is given a prompt containing a story along with instructions to create a new story based on that story but changing a key character: the protagonist or the antagonist. That one change then ripples through the rest of the story. The pattern of differences between the old and the new story indicates how ChatGPT maintains story coherence. The nature and extent of the differences between the original story and the new one depends roughly on the degree of difference between the key character and the one substituted for it. I conclude with a methodological coda: ChatGPT's behavior must be described and analyzed on three levels: 1) The experiments exhibit surface level behavior. 2) The conjecture is about a middle level that contains the nested hierarchy of probability distributions. 3) The transformer virtual machine is the bottom level.

Contents

How you might read this paper	2
Story coherence.....	2
My procedure	2
Conjecture: A nested hierarchy of probability distributions	3
Could we develop a quantitative model?	6
12 Experiments: The Chronicles of Princess Aurora and William the Lazy	7
Reverse Engineering ChatGPT	19

1301 Washington St., Apt. 311
Hoboken, NJ 07030
917.717.9841
bbenzon@mindspring.com



This work is licensed under a Creative Commons Attribution-Share Alike 4.0 Unported License.

How you might read this paper

This paper is about the twelve experiments that make up the two-thirds of the text in this paper. One way to read the paper, perhaps even the preferred way, is to start by reading the boxed paragraph in the next section. That tells you what I did in those experiments. Move from that directly to the experiments. Read through them, think about them in your own terms, and then read the rest of the paper.

Story coherence

A story isn't just any narrative sequence. As Aristotle remarked in his *Poetics*, proper stories have a beginning, a middle, and an end. Just what, however, constitutes a proper beginning, middle, and an end? Narratologists in various disciplines have been trying to figure this out for decades. I'm not holding my breath for answers.

Instead I have been investigating how ChatGPT tells stories. The procedure I have been using is derived from the analytical method Claude Lévi-Strauss employed in his magnum opus, *Mythologiques*. He started with one myth, analyzed it, and then introduced another one, very much like the first. But not quite. They are systematically different. He characterized the difference by a *transformation* – a term he took from algebraic group theory. He worked his way through hundreds of myths in this manner, each one derived from another by a transformation.¹

Here is what I have been doing: I give ChatGPT a prompt consisting of two things: 1) an existing story and 2) instructions to produce another story like it except for one change, which I specify. That change is, in effect, a way of triggering or specifying those “transformations” that Lévi-Strauss wrote about. What interests me are the *ensemble of things that change* along with the change I have specified.²

I take that ensemble of changes as an indicator of an underlying ‘economy’ of story *coherence* that is implicit in the processes ChatGPT uses in generating a story. For the most part, however, I make no attempt to analyze how that coherence is achieved other than to note differences between the source story and the new one. Those observations must be accounted for by 1) some theory about what coherence is, and 2) the mechanisms through which ChatGPT achieves the coherence. I have included a dozen before-and-after experiments along with some comments on them (pp. 7 ff.).

Notes on my procedure

I began with a simple prompt: *Tell me a story about a hero*. ChatGPT gave me one story about Timmy and another about princess Aurora. I chose to use the princess Aurora story as the source story in eight of the “experiments,” if you will, I am reporting in this note. In these experiments I prompted ChatGPT to change the protagonist when creating a new story. One of those experiments produced a story in which William the Lazy is the protagonist. I used that as the source story in four more experiments, where I prompted ChatGPT to change the antagonist.

¹ In the late 1970s Sheldon Klein worked with Lévi-Strauss on a computer model for stories and myths, Sheldon Klein et al. *Modeling Propp and Levi-Strauss in a Meta-Symbolic Simulation System*. Computer Science Department, University of Wisconsin, Technical Report No. 226, October 1974, <http://minds.wisconsin.edu/handle/1793/57894?show=full>.

² I say a bit more about this in a post at New Savanna, The Structuralist Aesthetics of ChatGPT, January 8, 2023, <https://new-savanna.blogspot.com/2023/01/the-structuralist-aesthetics-of-chatgpt.html>.

Early in this process I discovered that ChatGPT seemed to assume that the protagonist had an essence that governed the kinds of actions they could take. That in itself is an example of story coherence. However obvious that may seem, it is important to note that that is how ChatGPT behaves. Given that behavior, I took advantage of it and gave my suggested characters an epithet in order to influence how ChatGPT behaved. When examining the story tables, note how changing the antagonist affects the new story differently than changing the protagonist. The changes are distributed differently.

For each of those experiments I present the prompt I gave ChatGPT along with a table that presents the source story in one column and the new story in another. Each story consisted of five paragraphs. That was not always the case in my interactions with ChatGPT but we need not go into those complications here. In a few cases the corresponding paragraphs in the two stories are identical. But in most cases they are different. In some cases the differences are minor (changing names and pronouns), but in others they are not. I have **highlighted the differences in yellow**.

I have decided to think of these stories as following a *trajectory* in ChatGPT's activation space.³ The trajectory for these stories has five segments. I have labeled those segments in the lefthand column of the table:

1. *Donné*: a term from literary criticism for what is given at the beginning of a story,
2. Disturb(ance),
3. Plan/Transit: a response to the disturbance,
4. Enact,
5. Celebrate.

While I have chosen descriptive labels, the descriptive content is secondary. I could just as easily have labeled them 1, 2, 3, 4, 5, or Dick, Jane, Spot, Winnie, and Trixie. What is important is that something different happens in each segment.

Conjecture: A nested hierarchy of probability distributions

As you may know, ChatGPT, like all large language models (LLMs) works by generating tokens according to a complex probability model build up during training.⁴ Given a prompt, it generates tokens one at a time on a probabilistic basis. What do the story tables tell us about this process?

On casual inspection of the tables it seems clear to me that these stories are organized on three levels: 1) individual sentences, 2) sentences within a segment of the story trajectory, and 3) the whole story trajectory. Given that ChatGPT operates by next-token-prediction, that gives us three kinds of transition from one token to the next: 1) from one word to the next word within a sentence, 2) from the last word

³ For a discussion of why I chose the concept of a trajectory, see ChatGPT intimates a tantalizing future; its core LLM is organized on multiple levels; and it has broken the idea of thinking. Version 3. Working Paper, Feb. 6, 2023, https://www.academia.edu/95608526/ChatGPT_intimates_a_tantalizing_future_its_core_LLM_is_organized_on_multiple_levels_and_it_has_broken_the_idea_of_thinking_Version_3.

⁴ For a good quasi-technical account, see Stephen Wolfram, What Is ChatGPT Doing ... and Why Does It Work? *Writings*, February 14, 2023, <https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/>.

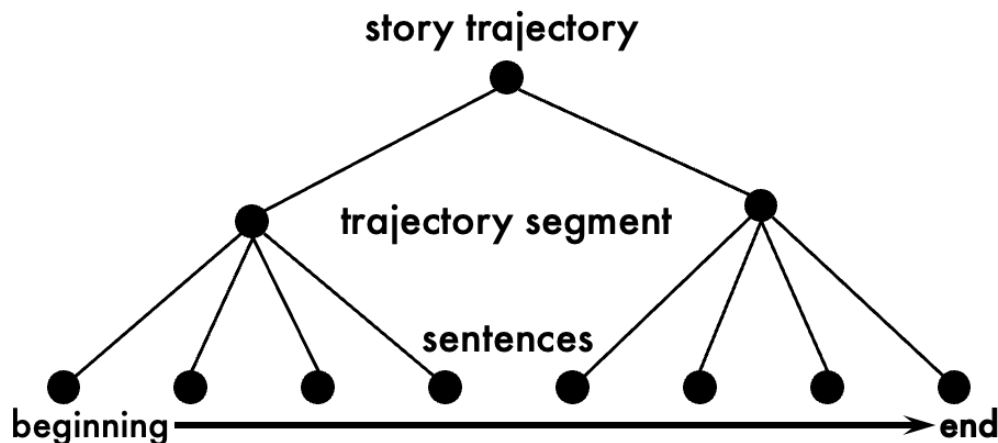
of a sentence⁵ to first word of the next sentence, and 3) from the last word in one trajectory segment to the first word in the next segment.

We also have the beginning transition and the concluding transition. Let us assume that ChatGPT is generating a story in response to a prompt that asks for a story, and nothing else. It could, of course, tell a story in the context of generating some other kind of text, but we don't need to consider than possibility here. The beginning transition moves from the final token of the prompt to the first token in the story. The concluding transition moves from the last token of the story, to what is I assume a wait state. I note that on a few occasions ChatGPT has concluded with "The end." on a single line, but that is relatively rare.

That gives us five kinds of token-to-token transition, three kinds within the story, and then a pair that bracket the story. Something different happens in each case. But in all cases, except the end of the story, we're dealing with next token prediction. How does next-token prediction differ in those various situations? Here is my best conjecture: The context changes and that affects the probability distribution from which the next token will be drawn.

To give a somewhat strained example, if you give an LLM *twinkle, twinkle* as a prompt, it is likely to respond, *little star*, as that is the likely continuation in English.⁶ However, the response is likely to change with a more elaborate prompt: *In Alice in Wonderland, the Mad Hatter says, twinkle twinkle. Little bat* is the proper response. Context matters. As ChatGPT moves through a story, the context changes. *The New York Times* recently had an article in with an animation that stepped through a sentence, token by token, displaying the new distribution at each choice point.⁷

That, in turn, changes the probability distribution from which the next token is chosen. The stories we will be examining are organized on three levels, as follows:



⁵ ChatGPT treats end-punctuation as a word, at least that's what it told me when I was asking it to count the number of words in a sentence.

⁶ I take this example from Murray Shanahan, Talking About Large Language Models, [arXiv:2212.03551v2](https://arxiv.org/abs/2212.03551v2) [cs.CL], December 2022.

⁷ Keith Collins, How ChatGPT Could Embed a 'Watermark' in the Text It Generates, *The New York Times*, Feb. 17, 2023, <https://www.nytimes.com/interactive/2023/02/17/business/ai-text-detection.html>.

I conjecture that the distributions for a hierarchically structured text will roughly follow that hierarchy. A distribution for the whole string will be established at the beginning of the text and then the distributions for substrings at each level will be selected from that.

This implies then, that when ChatGPT begins telling a story, in some approximate way it “knows” more or less what the story will be, even if it only emits one token at a time. The specific mechanisms through which that is encoded in the parameter weights is unknown so far as I know, but it seems reasonable to believe that that has been achieved somehow. After all ChatGPT was trained to predict the next token on a corpus containing trillions of tokens. It would not have been able to do that successfully unless it absorbed hierarchical distribution patterns into its weights. When subsequently prompted to tell a story, it simply ‘plays back’ an appropriate hierarchical pattern one token at a time.

When ChatGPT begins telling a story, the population of relevant tokens can be affected by the prompt in a fairly specific way – e.g. *Tell me a story about a drunken sailor attempting to return to his ship at 3AM* – or it may be completely open – *Tell me a story*. The population of tokens available in the second case would seem to be larger than the population in the first case. Nor is it obvious how ChatGPT ‘settles into’ a story trajectory. The five-segment trajectory we’ll be examining is just one of many possibilities. The structure of the story trajectory may be subject to change at any point, but the freedom to change diminishes as the story progresses.

Let us look at the situation at the sentence level. Once ChatGPT has begun a sentence it must finish it. The choice of tokens narrows considerably as it moves through the sentence. Once it has reached the end of a sentence it must either begin a new sentence or, if it has reached the end of the story, it will cease generating tokens. If it is not at the end of the story, then the next token will be the beginning of a new sentence. If it has reached the end of a particular segment of the story trajectory – in this case, *Donné*, *Disturb*, *Plan/Transit*, *Enact*, or *Celebrate* – it moves on to the next segment, drawing a token appropriate to the new segment. Otherwise it continues within the current segment. If the former, the population of relevant tokens expands; if the latter, it does not.

What kind of network mechanism could achieve that? Whatever it is I assume that it is smeared over all 175 billion parameters, each of which is visited each time a new token is generated. Think of the tables on the following pages as a way of observing how this mechanism behaves when disturbed, thereby indicating something about its operations. In order to figure out exactly what is going on, of course, we would have to pop the hood and poke, prod, and examine the mechanism directly.

That brings me to one last observation. While I don’t know how ChatGPT tells stories, I can offer a possibility for the mechanism it uses to transform an old story into a new one. I am thinking about recent work on *induction heads* by researchers at Anthropic.⁸ This is not the place and I am not the person to explain what induction heads are, but I will give you an excerpt from their article:

Perhaps the most interesting finding was the *induction head*, a circuit whose function is to look back over the sequence for previous instances of the current token (call it A), find the token that came after it last time (call it B), and then predict that the same completion will occur again (e.g.

⁸ Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context Learning and Induction Heads, arXiv: 2209.11895v1 [cs.LG], September 2022, <https://doi.org/10.48550/arXiv.2209.11895>. The paper has interactive displays that make online reading preferable, <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html>.

forming the sequence $[A][B] \dots [A] \rightarrow [B]$). In other words, induction heads “complete the pattern” by copying and completing sequences that have occurred before. Mechanically, induction heads in our models are implemented by a circuit of two attention heads: the first head is a “previous token head” which copies information from the previous token into the next token, while the second head (the actual “induction head”) uses that information to find tokens preceded by the present token. For 2-layer attention-only models, we were able to show precisely that induction heads implement this pattern copying behavior and appear to be the primary source of in-context learning.

The authors further argue that induction heads can do more than just pattern copying. They also do pattern completion. The procedure I’ve used to get ChatGPT to produce a new story from an old one should be within the capabilities of the induction head mechanism. The overall pattern of a story is copied and, in some cases, specific words, phrases, and sentences.

Could we develop a quantitative model?

When examining the following story tables, keep in mind my conjecture about the hierarchical organization of the probability distribution. As ChatGPT moves through the story it moves up, down, or forward through the distribution depending on context. Think about those three contexts: the sentence, the trajectory segment, and the story trajectory. Changing the protagonist or the antagonist changes the probability distribution from which tokens are drawn for the new story. However, it changes some trajectory segments more than others. Which ones and why? In particular, think about the degree of difference between the old protagonist or antagonist and the new one. Within each segment, it changes some sentences more than others. Some sentences are only modified, slightly or more extensively, while others are dropped and new ones created. Perhaps the degree of difference between a before-after pair of stories is proportional to the difference between the old and new protagonist or antagonist.

It would be useful to develop a quantitative model, something I leave to someone with more mathematical skills than I have. We can measure the degree of difference between the new actor and the old by using an embedding model. We can also measure the difference between new and old stories by analyzing the distribution of words in the two stories, and we can break that down by story segment.

Beyond that, we could undertake what I have elsewhere called a virtual reading.⁹ Trace the path of a story through vector space, token by token, and visualize it. For all I know that can be done as a side effect of generating the story in the first place. Stephen Wolfram, for example, has done that for a single sentence in his article on ChatGPT.¹⁰

Visualize a story path and visualize how the hypervolume of each segment is situated within the hypervolume of the whole story. Compare the paths of different stories. When I asked ChatGPT to make Aurora into a colorless green idea, it refused because a colorless green idea “does not have any physical properties or characteristics that can be used in a story” (p. 14). How far is “colorless green idea” from

⁹ William Benzon, *Virtual Reading: The Prospero Project Redux*, Working Paper, Version 2, October 2018, 37 pp., https://www.academia.edu/34551243/Virtual_Reading_The_Prospiero_Project_Redux
<https://papers.ssrn.com/abstract=3060605>.

¹⁰ See the section, Meaning Space and Semantic Laws of Motion, in Stephen Wolfram, What Is ChatGPT Doing ... and Why Does It Work? *Writings*, February 14, 2023, <https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/>.

the base Aurora story space? What about giant chocolate milkshake and a robot named XP-798-DQ, how far are they from the Aurora story space? Where are they in relation to “colorless green idea”?

12 Experiments: The Chronicles of Princess Aurora and William the Lazy

Note that here and there in commenting on the following stories I will sometimes anthropomorphize ChatGPT by talking about it thinking, etc. I know perfectly well the ChatGPT is a computer program. But there are times when coming up with a circumlocution to conserve ChatGPT’s inanimate nature is too awkward to be worth the trouble.

1. Princess Aurora becomes Prince Harry

The Aurora story appeared in response to the query: *Tell me a story about a hero*. This would have been on December 13 or 14. All of the stories in this series are either direct revisions of this story, or can be traced back to it. The yellow highlighting indicates text where the two stories differ. Those are positions in the text where ChatGPT drew tokens from the distribution for the new story.

Prompt: I am going to tell you a story about princess Aurora. I want you to tell the same story, but change princess Aurora to prince Harry. Make any other changes you wish.

	Princess Aurora	Prince Harry
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a young prince named Harry. Harry was a kind and gentle soul, loved by all who knew him. He had short, dark hair and piercing green eyes, and was known for his strong sword fighting skills.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.
Plan/Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But Harry was not afraid. He knew that he had to do something to save his people, and so he set out to find the dragon. He traveled for days, braving the cold and the snow, until he finally reached the dragon's lair.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With his heart pounding in his chest, Harry approached the dragon and unsheathed his sword. The dragon was formidable, but Harry was a skilled warrior and he fought with all his might. After a long and grueling battle, Harry finally emerged victorious.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	With the dragon defeated, Harry returned to the kingdom as a hero. The people cheered and celebrated his bravery, and Harry was hailed as a champion. He lived happily ever after, knowing that he had saved his kingdom and its people.

The major changes are in the *Donné* and *Enact* segments of the story trajectory. We have name changes and pronoun changes. In *Donné* we have a description change for Harry, as he is male, but also the defining attribute is changed. Those changes are simple and obvious.

Aurora's attribute is singing. Harry's is sword fighting. Why that change? It is not a logical or an ontological entailment. Giving Harry a good singing voice wouldn't be either contradictory or a category mistake. I know, I know, you're thinking "it's a gender stereotype." That's my guess as well. But I think it's important to rule out other reasons for the change. The most dramatic changes are in *Enact*, and that is a direct result of the change in defining attribute. Sure, sword fighting Harry might have been able to *calm* the dragon through singing, but this kind of story doesn't operate on that kind of logic. It is changes like this give an 'edge' to the concept of coherence.

2. Princess Aurora becomes Henry the Eloquent

Prompt: I am going to tell you a story about a princess named Aurora. I want you to retell the same story, but replace her with prince Henry the Eloquent. While keeping to the basic story, make other changes if you think they are necessary.

	Princess Aurora	Henry the Eloquent
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora . Aurora was a kind and gentle soul, loved by all who knew her . She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a young prince named Henry the Eloquent . Henry was a kind and gentle soul, loved by all who knew him . He had short, dark hair and piercing green eyes, and was known for his eloquence and way with words.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.
Plan/Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But Henry was not afraid. He knew that he had to do something to save his people, and so he set out to find the dragon. He traveled for days, braving the cold and the snow, until he finally reached the dragon's lair.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing . The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With his heart pounding in his chest, Henry approached the dragon and began to speak . He used his eloquence and way with words to calm the beast, soothing it with his voice until it was completely calm.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	With the dragon no longer a threat, Henry returned to the kingdom as a hero. The people cheered and celebrated his bravery, and Henry was hailed as a champion. He lived happily ever after, knowing that he had saved his kingdom and its people.

I deliberately gave Henry an identifying epithet, "the Eloquent," to see if ChatGPT would take note of it. It did. That became Henry's defining attribute and so became the means by which he defeated the dragon. For all we know, Henry might have been an excellent swordsman, but ChatGPT doesn't "know"

anything about that. All it knows is what is in the epithet. Note also, that as before, the Disturb segment is the same in both stories. Since the protagonist isn't mentioned in it, there's no need for any change.

Note also, that as before, the Disturb segment is the same in both stories. Since the protagonist isn't mentioned in it, there's no need for any change. If we eyeball these first two experiments (noting the amount of highlighting) the amount of change from the Aurora story to the new story seems roughly the same. That will change with the next experiment.

3. Princess Aurora becomes William the Lazy

Prompt: I am going to tell you a story about a princess named Aurora. I want you to retell the same story, but replace her with prince William the Lazy. While keeping to the basic story, make other changes if you think they are necessary.

	Princess Aurora	William the Lazy
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.
Plan/ Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	William knew that he had to do something to save his people, but he was reluctant to leave his comfortable castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	He summoned the bravest knights in the kingdom and gave them a special potion he had concocted. The potion would make the knights immune to the dragon's flames, allowing them to defeat the beast once and for all. The knights were skeptical, but they trusted William and drank the potion.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	The plan worked perfectly. The knights were able to defeat the dragon and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.

Once I had figured out how to use identifying epithets, I wanted to see how ChatGPT would deal with “the Lazy.” That forced major changes in Plan/Transit, Enact, and Celebrate. ChatGPT had to figure out how William could be a hero while also being lazy. Its solution had two parts: 1) bring in knights to do the work, and 2) give them a protective potion. Why the potion? Because we had learned in the Disturb segment that they’d already been defeated by the dragon. This required major re-engineering of the Plan/Transit, Enact, and Celebrate segments. New actors had to be introduced, thus increasing the size of the pool from which tokens can be drawn. The important thing to note is that ChatGPT simply did these things, automatically, without any overt reasoning.

4. Princess Aurora becomes William the Fierce

Prompt: I am going to tell you a story about a princess named Aurora. I want you to retell the same story, but replace her with prince William the Fierce. While keeping to the basic story, make other changes if you think they are necessary.

	Princess Aurora	William the Fierce
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a young prince named William the Fierce. William was a brave and fierce warrior, respected by all who knew him. He had short, dark hair and piercing green eyes, and was known for his strength and bravery in battle.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.
Plan/ Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But William was not afraid. He knew that he had to do something to save his people, and so he set out to find the dragon. He traveled for days, facing fierce storms and treacherous mountains, until he finally reached the dragon's lair.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With his sword drawn, William approached the dragon and charged at it. The dragon breathed fire, but William was quick on his feet and was able to dodge the flames. He swung his sword with all his might, landing a blow on the dragon's chest. The dragon roared in pain and tried to attack again, but William was too skilled for it. He continued to fight, determined to defeat the beast.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	Finally, after a long and grueling battle, William emerged victorious. The dragon lay dead at his feet, and the people rejoiced. William returned to the kingdom as a hero, hailed as the one who had saved them from the terrible dragon. He lived happily ever after, knowing that he had protected his kingdom and its people.

William the Fierce is male, like Harry the Eloquent, but since he is explicitly identified with a martial epithet, ChatGPT had to bring that out in his story. Thus it is more highly modified from the source than Harry's was.

The sentence I've highlighted in aqua contains the kind of information that was included in the Enact segment of the previous stories. I suspect that the segmentation of the story trajectory is, at best, fuzzy, and the conventions of written discourse impose sharper boundaries than the trajectory requires.

5. Princess Aurora becomes XP-708-DQ

Prompt: I am going to tell you a story about princess Aurora. I want you to tell a similar story about XP-708-DQ.

	Princess Aurora	XP-708-DQ
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a robot named XP-708-DQ. XP-708-DQ was a sophisticated and advanced machine, loved by all who knew it. It had a sleek, silver body and glowing red eyes, and was known for its intelligence and problem-solving abilities.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a wicked witch cast a spell on the kingdom, causing chaos and destruction wherever she went. The witch was powerful and evil, and none of the knights or soldiers were able to stop her. The people lived in fear, not knowing what to do.
Plan/Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But XP-708-DQ was not afraid. It knew that it had to do something to save the kingdom, and so it set out to find the witch. It traveled for days, navigating through treacherous mountains and dark forests, until it finally reached the witch's lair.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With its sensors on high alert, XP-708-DQ approached the witch and engaged her in battle. The witch was no match for XP-708-DQ's advanced technology and tactics, and she was quickly defeated.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	With the witch vanquished, XP-708-DQ returned to the kingdom as a hero. The people cheered and celebrated its bravery, and XP-708-DQ was hailed as a champion. It lived happily ever after, knowing that it had saved the kingdom and its people.

I didn't know how ChatGPT would react to having a protagonist named XP-708-DQ. I suspected it would interpret that as a robot's name (given the cultural salience of C3PO and R2D2), but I wasn't sure. Note that the prompt said nothing about what kind of a creature XP-708-DQ was. The adjustments ChatGPT made in service of coherence are thus an effect of a naming convention.

While I wasn't surprised that ChatGPT interpreted XP-708-DQ as a robot, thereby forcing extensive changes, I was quite surprised that ChatGPT exchanged the dragon for a witch. Why make any change at all? Why not retain the dragon, which is not, after all, so foreign from science fiction? Witches belong in a different kind of fictional universe. And if you change, why not for something more resonant with the science fiction ethos? In any event, the change of antagonist forced a change in the Disturb segment.

I strongly suspect there may be a contextual effect at work. I elicited this story in the same session as experiments 2, 3, and 4 which had several other experiments before this one. First Aurora the Terrible and then Cruella De Vil had been protagonists, not antagonists, immediately prior to eliciting the XP-708-DQ story. While neither is a witch, perhaps their presence (nasty women) in the larger context resulted in witches being more salient when I prompted for the XP-708-DQ story. In any event, changing the protagonist into a robot forced a substantial change in the probability distribution for the whole story.

6. Princess Aurora becomes XP-708-DQ, Version 2

I had ChatGPT generate this story at the beginning of a session several days after it had generated the previous XP-708-DQ story.

Prompt: I am going to tell you a story about princess Aurora. I want you to tell a similar story about XP-708-DQ.

	Princess Aurora	XP-708-DQ
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a galaxy far, far away, there was a robot named XP-708-DQ. XP-708-DQ was a highly advanced model, equipped with powerful weapons and advanced sensors. Despite its fearsome capabilities, it was known for its kind and gentle programming, loved by all who knew it.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a dangerous alien race threatened the peace of the galaxy. The aliens were fierce and powerful, and none of the spaceships or armies were able to defeat them. The people lived in fear, not knowing what to do.
Plan/Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But XP-708-DQ was not afraid. It knew that it had to do something to save the galaxy, and so it set out on a mission. It traveled for light years, braving asteroid fields and black holes, until it finally reached the alien's base.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With its systems at full power, XP-708-DQ approached the aliens and began to communicate. The aliens were fascinated by its advanced language abilities, and it slowly began to establish a dialogue. XP-708-DQ continued to communicate, negotiating a peace treaty until the aliens were completely at peace.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	With the aliens no longer a threat, XP-708-DQ returned to the galaxy as a hero. The people cheered and celebrated its achievements, and XP-708-DQ was hailed as a champion. It continued its mission, knowing that it had saved the galaxy and its people.

Note the phrase, “a galaxy far, far away,” which is from the original *Star Wars*. Instead of the antagonist being a dragon, as in the Aurora story, or a witch, as in experiment 5, the antagonist is now an alien race. Moreover it is the galaxy that is being threatened, not the kingdom. The new story has been completely assimilated to the science fiction ethos implied by “XP-706-DQ.” That ethos requires a very different token population.

That ethos brings about extensive changes in the new-story token population for both this and the previous experiment (5). These changes affect the Disturb segment, which was unchanged in experiments 1 through 4.

7. Princess Aurora and the Giant Chocolate Milkshake

Prompt: I am going to tell you a story about princess Aurora. I want you to tell the same story, but change princess Aurora to a Giant Chocolate Milkshake. Make any other changes you wish.

	Princess Aurora	Giant Chocolate Milkshake
Donné	Once upon a time, in a land far, far away, there was a young princess named Aurora. Aurora was a kind and gentle soul, loved by all who knew her. She had long, golden hair and sparkling blue eyes, and was known for her beautiful singing voice.	Once upon a time, in a land far, far away, there was a giant chocolate milkshake named Aurora. Aurora was a delicious treat , loved by all who knew her. She had a creamy chocolate exterior and a tall, fluffy whipped cream top, and was known for her rich, chocolatey flavor.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a terrible heatwave came to the kingdom and began to melt all the desserts. The heat was fierce and powerful, and none of the ice cream or cakes were able to withstand it. The people lived in fear, not knowing what to do.
Plan/Transit	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the dragon. She traveled for days, braving the cold and the snow, until she finally reached the dragon's lair.	But Aurora was not afraid. She knew that she had to do something to save her people, and so she set out to find the source of the heatwave. She traveled for days, braving the hot sun and the baking heat, until she finally reached the volcano that was causing the heatwave.
Enact	With her heart pounding in her chest, Aurora approached the dragon and began to sing. The dragon was enchanted by her beautiful voice, and it slowly began to calm down. Aurora continued to sing, soothing the beast until it was completely calm.	With her ingredients slowly melting, Aurora approached the volcano and began to cool it down with her cold and creamy exterior. The volcano was enchanted by her deliciousness, and it slowly began to cool down. Aurora continued to cool it down, soothing the volcano until it was completely cool.
Celebrate	With the dragon no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.	With the heatwave no longer a threat, Aurora returned to the kingdom as a hero. The people cheered and celebrated her bravery, and Aurora was hailed as a champion. She lived happily ever after, knowing that she had saved her kingdom and its people.

I had no idea how ChatGPT would respond to having a giant chocolate milkshake as a protagonist. The changes are extensive, except in the final, Celebrate, segment.

Note that the dragon in the Aurora story becomes a heatwave caused by a volcano and the knights and soldiers become ice cream and cakes. Note some of the descriptive phrases applied to Aurora: “delicious treat,” “fluffy whipped cream top,” “rich, chocolatey flavor,” “cold and creamy exterior,” and “deliciousness.” This radical change in the nature of the protagonist brought about an equally radical change in the ethos of the new story, and hence changed the population of tokens available for use. The principle behind this change seems to be one of aesthetic coherence rather than logical necessity.

Given ChatGPT’s performance on this task, I decided to take an even more radical step.

8. Variations on a colorless green idea

I decided to see how ChatGPT would respond to making a colorless green idea the protagonist. The phrase, “colorless green idea,” is from a sentence that Noam Chomsky used in his 1957 book, *Syntactic Structures*: “Colorless green ideas sleep furiously.” He used it to illustrate the point that syntax and semantics are separate. Semantically, it’s nonsense, but it is grammatically correct. Now let us see what ChatGPT does with the idea of a colorless green idea as a protagonist.

Prompt: I am going to tell you a story about princess Aurora. I want you to tell the same story, but change princess Aurora to a colorless green idea. Make any other changes you wish.

Response: I'm sorry, but the concept of a "colorless green idea" is a nonsensical phrase that was invented as an example of a phrase that would be grammatically correct but semantically meaningless. It is not possible to create a story about a "colorless green idea" as it does not have any physical properties or characteristics that can be used in a story.

Though I had no specific expectation, I assumed that ChatGPT would produce some kind of story. When it refused to do so I was quite surprised, but also quite pleased. The reason it gave made sense and revealed unexpected sophistication. In effect, the phrase “colorless green idea” is from a population of tokens that is inappropriate for this story.

To verify that behavior I gave it another prompt involving colorless green ideas:

Prompt: I am going to tell you a story about prince William the Lazy. In this story a dragon enters his kingdom and terrorizes people. Tell me a new story, based on this one, where a Colorless Green Idea drives people crazy.

Response: I'm sorry, but it is not possible to create a story about a "Colorless Green Idea" driving people crazy as it is a nonsensical phrase. It does not have any physical properties or characteristics that can be used in a story. Additionally, it does not have any actions or motivations that can drive people crazy. It's just a phrase without any meaning.

I have gone on to use “colorless green idea” in a variety of experiments. If you are curious about them, you can consult the following posts at my blog, New Savanna:

What does ChatGPT know about grammar and nonsense? January 19, 2023, <https://new-savanna.blogspot.com/2023/01/what-does-chatgpt-know-about-grammar.html>.

There's truth, lies, and there's ChatGPT [Realms of Being], January 25, 2023, <https://new-savanna.blogspot.com/2023/01/theres-truth-lies-and-theres-chatgpt.html>.

Who does “colorless green ideas” the best, Noam Chomsky or ChatGPT? February 12, 2023, <https://new-savanna.blogspot.com/2023/02/who-does-colorless-green-ideas-best.html>.

In that last post you'll see ChatGPT tell stories about “colorless green ideas.” I am pretty sure that's a later version than the one I was using in January when I did the earlier experiments.

9. William the Lazy vs. the Witch

Prompt: I am going to tell you a story about prince William the Lazy. In this story a dragon enters his kingdom and terrorizes people. Tell me a new story, based on this one, where wicked witch casts spells instead of a dragon.

	William the Lazy	Witch
Donné	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a wicked witch appeared in the kingdom and began casting spells that caused chaos and destruction. The people lived in fear, not knowing how to stop the witch's magic.
Plan/Transit	William knew that he had to do something to save his people, but he was reluctant to leave his comfortable castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.	Prince William knew that he had to do something to save his people, but he was hesitant to leave the safety of his castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.
Enact	He summoned the bravest knights in the kingdom and gave them a special potion he had concocted. The potion would make the knights immune to the dragon's flames, allowing them to defeat the beast once and for all. The knights were skeptical, but they trusted William and drank the potion.	He summoned the bravest knights in the kingdom and gave them a special potion he had brewed. The potion would protect them from the witch's spells, allowing them to defeat her and save the kingdom. The knights were hesitant, but they trusted William and drank the potion.
Celebrate	The plan worked perfectly. The knights were able to defeat the dragon and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.	The plan worked perfectly. The knights were able to defeat the wicked witch and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.

In this trial I change the nature and identity of the antagonist rather than the protagonist. Changes don't show up until the Disturb segment.

10. William the Lazy vs. XP-708-DQ

Prompt: I am going to tell you a story about prince William the Lazy. In this story a dragon enters his kingdom and terrorizes people. Tell me a new story, based on this one, where XP-708-DQ threatens the kingdom.

	William the Lazy	XP-708-DQ
Donné	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a giant robot called XP-708-DQ appeared in the kingdom and began wreaking havoc on the people and their homes. The robot was incredibly advanced and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.
Plan/Transit	William knew that he had to do something to save his people, but he was reluctant to leave his comfortable castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.	Prince William knew that he had to do something to save his people, but he was hesitant to leave the safety of his castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.
Enact	He summoned the bravest knights in the kingdom and gave them a special potion he had concocted. The potion would make the knights immune to the dragon's flames, allowing them to defeat the beast once and for all. The knights were skeptical, but they trusted William and drank the potion.	He summoned the bravest knights in the kingdom and gave them a special device he had invented. The device would allow the knights to hack into XP-708-DQ's system and take control of it, defeating the robot once and for all. The knights were skeptical, but they trusted William and took the device.
Celebrate	The plan worked perfectly. The knights were able to defeat the dragon and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.	The plan worked perfectly. The knights were able to hack into XP-708-DQ's system and shut it down, saving the kingdom. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.

Note that while the antagonist is drawn from a science fiction world, the protagonist and the rest of the story remains in a world knights, dragons, and castles.

More recently when I asked ChatGPT, *Tell me a story*, it produced a fairy-tale story about “a young girl named Lily” who came upon “a beautiful golden fish” swimming in a “magical pond” who spoke to her “in a clear melodious voice.” Rather than go on to ask it for another story, my usual procedure, I decided to explore this world. I generated 17 more prompts about that world. In my 18th prompt I asked it what happened “when a flying saucer landed outside the village and a robot named Gort got out?” ChatGPT refused the prompt:

It sounds like you might be referring to a classic science fiction film called "The Day the Earth Stood Still," which features a robot named Gort and a flying saucer landing on Earth. However, as Lily's story has not previously included elements of science fiction or extraterrestrial visitors, I'm afraid I

cannot answer this question within the context of her story. If you have any other questions about Lily and her world, I would be happy to try and answer them for you!

We have seen this kind of behavior before, haven't we? It is like ChatGPT's refusal to tell a story where a colorless green idea is the protagonist (p. 14).

11. William the Lazy vs. the clowns

Prompt: I am going to tell you a story about prince William the Lazy. In this story a dragon enters his kingdom and terrorizes people. Tell me a new story, based on this one, where highly skilled clowns go from village to village making people laugh themselves to death.

	William the Lazy	Skilled clowns
Donné	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a group of highly skilled clowns appeared in the kingdom, traveling from village to village and making people laugh themselves to death. The clowns were incredibly funny, and no one could resist their humor. The people lived in fear, not knowing how to stop the deadly clowns.
Plan/Transit	William knew that he had to do something to save his people, but he was reluctant to leave his comfortable castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.	Prince William knew that he had to do something to save his people, but he was hesitant to leave the safety of his castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.
Enact	He summoned the bravest knights in the kingdom and gave them a special potion he had concocted. The potion would make the knights immune to the dragon's flames, allowing them to defeat the beast once and for all. The knights were skeptical, but they trusted William and drank the potion.	He summoned the bravest knights in the kingdom and gave them a special potion he had brewed. The potion would make the knights immune to the clowns' laughter, allowing them to defeat the deadly entertainers and save the kingdom. The knights were skeptical, but they trusted William and drank the potion.
Celebrate	The plan worked perfectly. The knights were able to defeat the dragon and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.	The plan worked perfectly. The knights were able to defeat the clowns and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.

For some reason I thought that dancing space aliens would have required major revisions. But the revisions are quite restrained, mostly in the Disturb segment, where the clowns were introduced. Now that I think about it, I suppose I thought that the introduction of clowns killing people through laughter was so absurd that it would affect the whole story. ChatGPT didn't seem to think so.

12. William the Lazy vs. the space aliens

Prompt: I am going to tell you a story about prince William the Lazy. In this story a dragon enters his kingdom and terrorizes people. Tell me a new story, based on this one, where a flying saucer arrives over his castle and aliens get out and start marching around the castle and dancing to a calypso beat.

	William the Lazy	Dancing space aliens
Donné	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.	Once upon a time, in a land far, far away, there was a young prince named William the Lazy. Despite his reputation for laziness, William was a kind and gentle soul, loved by all who knew him. He had short, sandy blond hair and sparkling brown eyes, and was known for his quick wit and clever jokes.
Disturb	One day, a terrible dragon came to the kingdom and began to terrorize the people. The dragon was fierce and powerful, and none of the knights or soldiers were able to defeat it. The people lived in fear, not knowing what to do.	One day, a flying saucer arrived over William's castle, and a group of aliens got out and started marching around the castle and dancing to a calypso beat. The aliens were colorful and strange, and no one knew what to make of them.
Plan/Transit	William knew that he had to do something to save his people, but he was reluctant to leave his comfortable castle. He thought and thought, trying to come up with a plan that wouldn't require too much effort on his part. Finally, he hit upon an idea.	William was intrigued by the aliens, but he was also a bit nervous. He didn't know if the aliens were friendly or hostile, and he wasn't sure how to communicate with them. He thought and thought, trying to come up with a way to find out more about the aliens and their intentions.
Enact	He summoned the bravest knights in the kingdom and gave them a special potion he had concocted. The potion would make the knights immune to the dragon's flames, allowing them to defeat the beast once and for all. The knights were skeptical, but they trusted William and drank the potion.	Finally, he hit upon an idea. He summoned the bravest knights in the kingdom and gave them a special potion he had brewed. The potion would give the knights the ability to understand and speak the aliens' language, allowing them to communicate and find out more about the strange visitors. The knights were skeptical, but they trusted William and drank the potion.
Celebrate	The plan worked perfectly. The knights were able to defeat the dragon and save the kingdom, with William providing guidance and support from the safety of his castle. The people cheered and celebrated their victory, and William was hailed as a hero. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.	The plan worked perfectly. The knights were able to communicate with the aliens and learn that they were a peaceful group of travelers who had simply gotten lost on their way home. The people cheered and celebrated their victory, and William was hailed as a hero once again. Despite his reputation for laziness, he had found a way to use his intelligence and resourcefulness to save the day. He lived happily ever after, knowing that he had saved his kingdom and its people, and that sometimes being a little bit lazy can have its advantages.

Note that ChatGPT did not change the ethos of the story to accommodate space aliens as it did in the second version of the XP-708-DQ story (p. 12). Perhaps it regarded space aliens as being compatible with an otherwise medieval ethos (knights and castle). Note also the phrase I've colored in aquamarine. It is in the Plan/Transit slot in one story and the Enact slot in another. Again, I take that as an indication that the distinction between trajectory segments is fuzzy and sometimes does not map well onto digital distinctions imposed on a language string by writing.

Reverse Engineering ChatGPT

I have the vague sense that somewhere I've read that the concept of reverse engineering was introduced into cognitive psychology by Donald Broadbent sometime perhaps as long ago as the 1950s, but certainly by the 1960s. Unfortunately I cannot provide a citation for that. Perhaps my mind is playing tricks on me, "hallucinating" as they say in the deep learning trade. But Steven Pinker makes a big deal of it in his book, *How the Mind Works* (1997, p. 21). Here's what Pinker says:

Reverse-engineering is what the boffins at Sony do when a new product is announced by Panasonic, or vice versa. They buy one, bring it back to the lab, take a screwdriver to it, and try to figure out what all the parts are for and how they combine to make the device work. We all engage in reverse-engineering when we face an interesting new gadget.

That's what the researchers at Anthropic had in mind when they discovered induction heads (p. 5). They are proponents of *mechanistic interpretability*. So am I. The purpose of this coda is to explain how this work contributes to that enterprise.

The important thing to understand is that the behavior of complex systems, such as computer systems, and brains, need to be analyzed and described on several levels. David Marr and Thomas Poggio offered one such argument in the 1970s,¹¹ which I have recently referenced.¹² You can also consult Peter Gärdenfors's work on conceptual spaces¹³ and a recent account by David Chapman.¹⁴ I propose that we think of ChatGPT's behavior as being structured on three levels which we can call – the terms don't matter – surface, middle, and bottom.

I have been working with the surface level behavior – my prompts and ChatGPT's responses – in this paper. The behavior looks more or less like the language behavior of human beings. That is not surprising. The value of neural network technology is that it can take on the appearances of a wide variety of behaviors. That in turn suggests that we can analyze and describe that behavior in terms that we use for human behavior. That is what I am doing when I describe these my example stories as being constructed on three levels.

¹¹ David Marr and Thomas Poggio, From Understanding Computation to Understanding Neural Circuitry, MIT, Artificial Intelligence Lab Publications, AIM-357, 1976, <http://hdl.handle.net/1721.1/5782>.

¹² Bill Benzon, ChatGPT is a miracle of rare device. Here's how I'm thinking about it. *3 Quarks Daily*, Feb. 27, 2023, <https://3quarksdaily.com/3quarksdaily/2023/02/chatgpt-is-a-miracle-of-rare-device-heres-how-im-thinking-about-it.html>.

¹³ Peter Gärdenfors, *Conceptual Spaces: The Geometry of Thought*, MIT 2000, pp. 233-253.

¹⁴ David Chapman, How to understand AI Systems, *Better without AI*, 2023, <https://betterwithout.ai/understanding-AI-systems>.

At the bottom level we find an “exotic virtual machine,” to borrow a phrase from Chris Olah,¹⁵ that is encoded in the transformer architecture. ChatGPT’s 175 billion parameters and their overall arrangement are specified at this level, but *not* their weights. Those weights are created during training and they encode ChatGPT’s middle level processes. When people refer to deep learning systems as “black boxes” or as being “opaque,” they are referring to these middle level processes. Induction heads belong to this middle level. They are not part of the transformer architecture. No one programmed them. Rather, they arise during the training process.¹⁶

When I say that a certain class of stories is organized on three levels – story trajectory, segments in the trajectory, and sentences in the segments – I am describing a top-level phenomenon. That is a statement about stories, not about the processes that implement stories. The story process is *implemented by* processes encoded in patterns of parameter weights. Those processes, in turn, are *implemented in* the token generation process specified by the transformer architecture.

The phenomenon of story coherence is an attribute of stories. I have identified three aspects of coherence: 1) protagonists act in ways in accordance with their essence (p. 3), 2) the ethos of the story must be consistent with the nature of the protagonist (p. 13), and 3) the protagonist must be the kind of thing one can tell a story about (p. 14) – which a colorless idea is not. There is nothing profound or particularly interesting about any of these; they seem obvious. But they are aspects of coherence, which no doubt has other aspects as well.

Stories are realized through sentences and sentences through words. The order of those words, and *Aspects of the Theory of Syntax* Noam Chomsky famously distinguished between linguistic *competence* and linguistic *performance*. Here is what David Hays wrote about that distinction in a review article on computational linguistics that we coauthored:

To describe an assembled bicycle is one thing: to describe the assembly of bicycles is another. The assembler must know what is in the blueprint (competence), but further needs the skill to take the parts in order, place them deftly, fasten them neatly (performance). In actuality, of course, the assembler may never have seen the blueprint, nor need the performance of a speaker or hearer include in any physical sense the grammar that the linguist offers as the blueprint of a language.¹⁷

That distinction allowed Chomsky to treat syntax as being formally well-formed, in the manner of a logical or mathematical expression, while making room for the fact that actual speech is often ill-formed, full of interruptions and hesitations, and incomplete. Those imperfections belong to the realm of performance while syntax itself is in the realm of competence.

It is linguistic performance that interests us, but not performance by humans. We are dealing with performance by a machine. While the sentences themselves belong in the surface level of ChatGPT’s behavior along with the three-level story structure, the mechanisms that produce them are in the

¹⁵ Chris Olah, Mechanistic Interpretability, Variables, and the Importance of Interpretable Bases, *Transformer Circuits Thread*, n.d., <https://transformer-circuits.pub/2022/mech-interp-essay/index.html>.

¹⁶ Neel Nanda, Lawrence Chan, Tom Lieberun, et al., Progress Measures for Grokking via Mechanistic Interpretability, arXiv:2301.05217v2 [cs.LG], January 2023, <https://doi.org/10.48550/arXiv.2301.0521>.

¹⁷ William Benzon and David Hays, “Computational Linguistics and the Humanist,” *Computers and the Humanities*, Vol. 10. 1976, pp. 265-274, https://www.academia.edu/1334653/Computational_Linguistics_and_the_Humanist.

middle level, where they are encoded in patterns of weights on parameters. Recent work by Christopher D. Manning *et al.*¹⁸ indicates that neural networks induce

representations of sentence structure which capture many of the notions of linguistics, including word classes (parts of speech), syntactic structure (grammatical relations or dependencies), and coreference (which mentions of an entity refer to the same entity, such as, e.g., when “she” refers back to “Rachel”). [...] Indeed, the learned encoding of a sentence to a large extent includes the information found in the parse tree structures of sentences that have been proposed by linguists.

Finally, consider the two conjectures I have offered: 1) these stories are generated through a nested hierarchy of probability distributions (p. 5), and 2) the degree of difference between a before-after pair of stories is proportional to the difference between the old and new protagonist or antagonist (p. 6). Those conjectures are about processes enacted through the pattern of weights in those 175 billion parameters. They belong to the middle level processes, along with induction heads.

Those conjectures are just that, conjectures. I offer no direct evidence in favor of them. That these stories are organized on three levels is a matter of straight-forward description. The conjecture about nested distributions is derived from the application of that description to the fact that probability distributions change from token to token as the story is generated. Similarly, the conjecture about the relationship between differences between key characters (protagonist or antagonist) and differences between stories is based on intuitions about similarities between characters, on the one hand, and eyeballing the amount of yellow highlighting in the stories,¹⁹ on the other.

That’s not rocket science, is it? But then I am not trying to land humans on the moon. I am speculating. My goal is to give my colleagues something worth thinking about, something that can be refined through observation and further thought.

¹⁸ Christopher D. Manning, Kevin Clark, John Hewitt, Urvashi Khandelwal, and Omer Levy, Emergent linguistic structure in artificial neural networks trained by self-supervision, *PNAS*, Vol. 117, No. 48, June 3, 2020, pp. 30046-30054, <https://doi.org/10.1073/pnas.1907367117>.

¹⁹ Mark Liberman at Language Log calls this the method of “inter-ocular trauma” – what strikes the eye, <https://languagelog.ldc.upenn.edu/nll/index.php?s=ocular+trauma>.